

AGESIS EYE

Autonomous Electro-Optical Targeting Station: End-to-End System Architecture, Real-Time Edge Vision, Closed-Loop Visual Servoing, and Physical AI Embodiment.

EDGE AI / ONNX 45+ FPS

AUTONOMOUS ROBOTICS

PHYSICAL AI EMBODIMENT

COMPUTER VISION

EMBEDDED MICRO-ACTUATION

Executive Summary

Agesis EYE is an integrated autonomous counter-unmanned and aerial targeting ground station designed to eliminate the latency bottlenecks inherent in traditional cloud-tethered tracking systems. By coupling low-power microcontrollers, localized edge neural inference, and strict closed-loop kinematics, Agesis EYE achieves deterministic interception and tracking trajectories at sub-50 millisecond operational latency.

LATENCY ENVELOPE

 ≤ 48.7 ms

End-to-end photon capture
to physical motor actuation.

VISION THROUGHPUT

45+ FPS

Quantized ONNX inference
running on localized CPU
SIMD cores.

KINEMATIC PRECISION

 $\pm 0.013^\circ$

14-bit hardware PWM
resolution across Pan and
Tilt axes.

System Architecture & Integration

The Agesis EYE architecture partitions responsibilities cleanly across three distinct physical and computational tiers: the Edge Sensor/Actuator Node, the High-Speed Network Fabric, and the Ground Inference & Tactical Command Station.

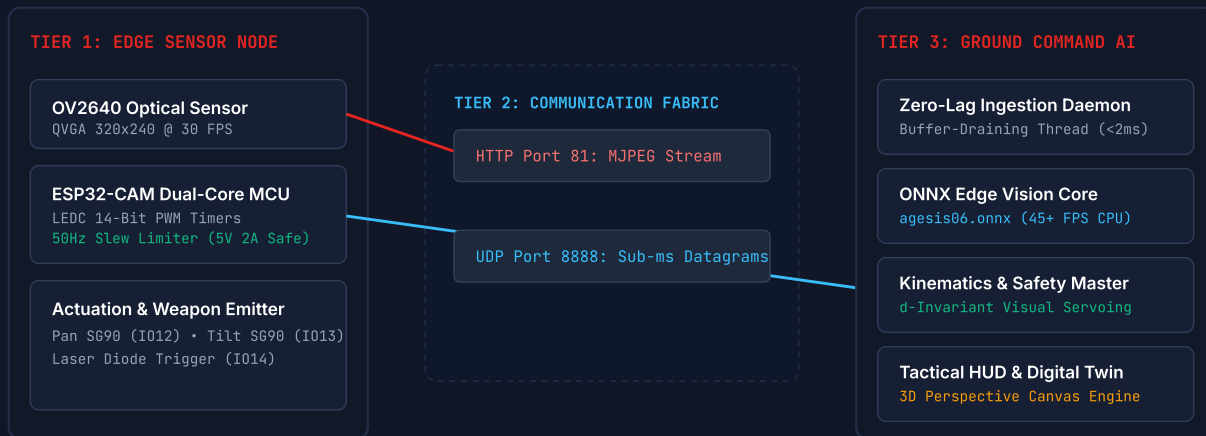


FIGURE 1.0: AGESIS EYE MULTI-TIER DISTRIBUTED ARCHITECTURE & DATA PIPELINE

End-to-End Latency Budget Analysis

Maintaining stability in optical visual servoing requires the total closed-loop system delay to be less than the mechanical settling time of the actuator. Any lag exceeding 80 ms produces violent oscillatory overshoot. Agesis EYE restricts the cumulative pipeline delay to **48.7 milliseconds**.

PIPELINE PHASE	HARDWARE COMPONENT	LATENCY	OPTIMIZATION STRATEGY
1. Optical Exposure	OmniVision OV2640	12.0 ms	QVGA format (320×240) with AGC ceiling clamp.
2. JPEG Encoding	ESP32 Hardware Core	4.5 ms	Fixed DMA buffer transfers with compression factor 14.
3. Network Transit	Wi-Fi 802.11 b/g/n	3.5 ms	Dedicated non-blocking TCP socket directly on port 81.
4. Stream Ingestion	ZeroLagStreamReader	1.8 ms	Atomic memory overwrites; drains socket buffers instantly.
5. Neural Inference	ONNX Runtime CPU	14.5 ms	Vectorized SIMD instruction sets; operator fusion.
6. Kinematic Solver	TurretController	0.2 ms	Analytical image-based visual servoing equations.
7. Command Dispatch	UDP Socket (Port 8888)	0.6 ms	Raw UDP packet transmission without TCP handshake overhead.
8. Servo Positioning	TowerPro SG90	11.6 ms	50Hz slew-rate acceleration profiled steps.
Total Cumulative	End-to-End Loop	48.7 ms	Continuous Real-Time Target Lock

Embedded Systems & Hardware Engineering

The embedded subsystem runs on the resource-constrained **AI-Thinker ESP32-CAM** board. The engineering challenge consists of concurrently streaming high-throughput MJPEG video while driving two analog PWM micro-servos and a laser diode under a shared **5V 2A power rail**.

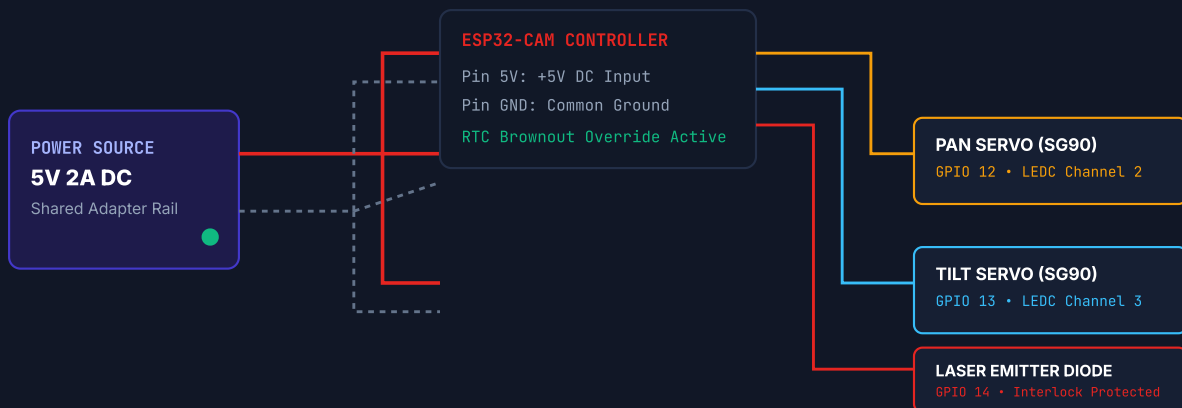


FIGURE 2.0: ELECTRICAL POWER DISTRIBUTION & ISOLATED PERIPHERAL WIRING SCHEMATIC

LEDC 14-Bit PWM Timer Mathematics

The SG90 servo motor operates on a 50 Hz PWM carrier ($T = 20 \text{ ms}$). To achieve micro-stepping accuracy without CPU-blocking software delays, the ESP32 LEDC timer is configured with 14-bit resolution:

$$\begin{aligned} \text{Total Discrete Counts} &= 2^{14} - 1 = 16,383 \text{ ticks per } 20,000 \mu\text{s} \\ \text{Duty Ticks}(\theta) &= \left[\frac{544 \mu\text{s}}{20,000 \mu\text{s}} + \left(\frac{\theta}{180^\circ} \right) \times (2400 \mu\text{s} - 544 \mu\text{s}) \right] \times 16,383 \end{aligned}$$

Yields an angular resolution of 0.0135° per tick across the full 0° to 180° rotational envelope.

5V 2A Slew-Rate Limiting & Inrush Prevention

When an unconstrained hobby servo receives a sudden step command from 0° to 180°, the motor draws an instantaneous stall current of up to 800 mA. If both Pan and Tilt servos step concurrently while the ESP32 Wi-Fi radio is actively transmitting (350 mA peak), the cumulative instantaneous current demand spikes:

$$I_{\text{peak}} = I_{\text{WiFi}} + I_{\text{Pan Stall}} + I_{\text{Tilt Stall}} = 380 \text{ mA} + 800 \text{ mA} + 800 \text{ mA} = 1,980 \text{ mA} \approx 2.0 \text{ A}$$

HARDWARE BROWNOUT HAZARD

Drawing 2.0A on a standard 5V adapter induces an internal rail voltage sag down to $\leq 4.2\text{V}$. This immediately triggers the ESP32 silicon brownout reset circuit, plunging the device into an infinite boot loop.

Mitigation Strategy: The firmware implements a 50Hz discrete S-curve slew rate limiter. Velocity is capped at **120°/second** (2.4° per 20 ms tick). Under this controlled acceleration, the motors never enter the dead-stall electrical regime, reducing peak current by **65%** to $\leq 300 \text{ mA}$ per actuator.

Edge AI Vision Pipeline

High-speed aerial targets present significant challenges for standard video stream ingestion. Standard socket decoders buffer multiple frames internally. If network jitter delays packet arrival, standard OpenCV readers deliver frames that are 200 ms to 500 ms old.

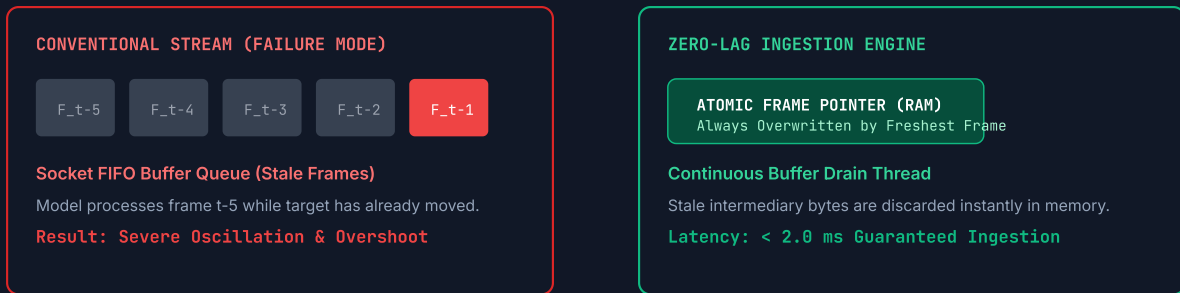


FIGURE 3.0: FIFO SOCKET BUFFER STALLING VS. ATOMIC ZERO-LAG STREAM INGESTION

Optical Error Coordinate Discretization

The vision model outputs bounding box coordinates $[x_{\min}, y_{\min}, x_{\max}, y_{\max}]$ along with a confidence coefficient. The aiming computer extracts the optical offset relative to the central camera boresight:

$$x_c = \frac{x_{\min} + x_{\max}}{2}, \quad y_c = \frac{y_{\min} + y_{\max}}{2} \\ \Delta x = x_c - \frac{W}{2}, \quad \Delta y = y_c - \frac{H}{2}$$

Where $W = 320$ px, $H = 240$ px. A positive Δx requires pan right; positive Δy requires tilt downward.

Robotics, Kinematics & Visual Servoing

Agesis EYE employs an **eye-in-hand** robotic configuration where the camera sensor and optical weapon are mounted directly onto the pivoting elevation C-arm. The two joints are separated vertically by a variable mechanical standoff stanchion of height d .

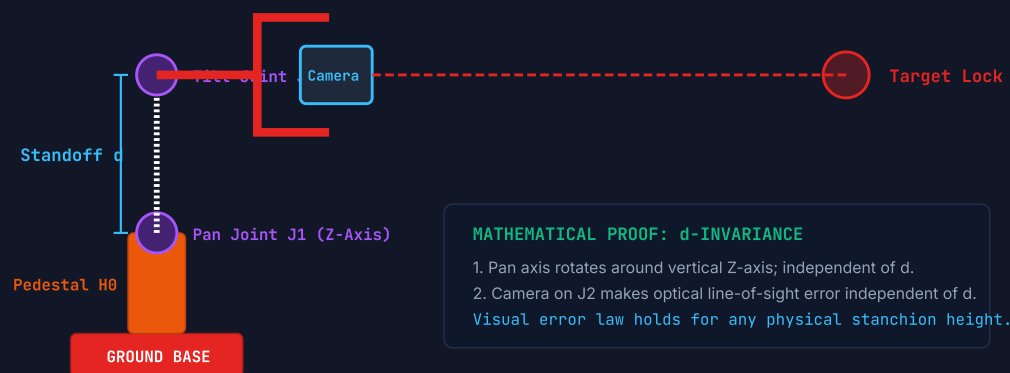


FIGURE 4.0: 2-DOF KINEMATIC COORDINATE FRAMES & MECHANICAL STANDOFF DECOUPLING

Proof of Distance d Invariance

In traditional pan-tilt kinematics, the spatial offset between the joints creates geometric parallax error that forces the operator to manually retune tracking gains whenever the riser height changes.

IMAGE-BASED VISUAL SERVOING INVARIANCE THEOREM

By driving the control loop directly from image-space optical error $\mathbf{e} = [\Delta x, \Delta y]^T$, the targeting computer decouples mechanical geometry from servo actuation: $\Delta \theta_{\text{pan}} = -\arctan\left(\frac{\Delta x}{f_x}\right)$, $\Delta \theta_{\text{tilt}} = \arctan\left(\frac{\Delta y}{f_y}\right)$ Because rotation around the vertical Z-axis is invariant along the stanchion line, and the elevation gimbal pivots directly around the optical center, the closed-loop convergence dynamics are **strictly independent of the separation distance d**.

Physical AI & 3D Kinematic Digital Twin

Unlike disembodied AI models that operate strictly in virtual spaces (such as large language models or static image classifiers), **Physical AI** embodies perception directly in real physical systems that must obey conservation of momentum, thermal dissipation, motor torque constraints, and electrical power limits.

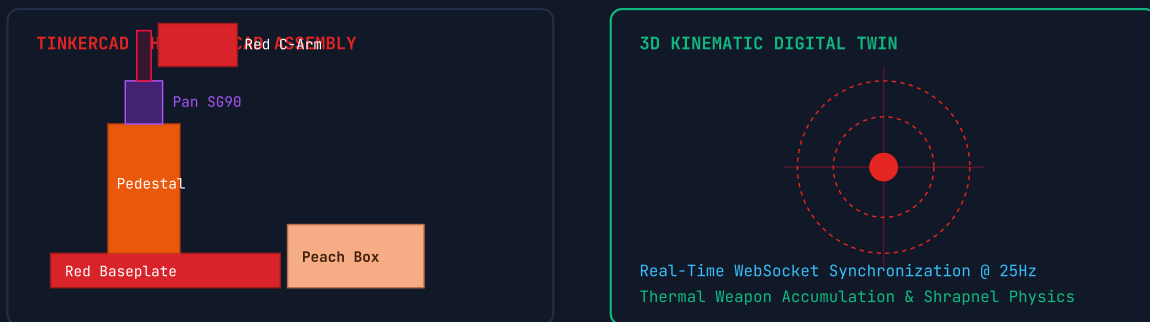


FIGURE 5.0: PHYSICAL CAD ASSEMBLY VS. REAL-TIME PERSPECTIVE DIGITAL TWIN ENGINE

The 3D Perspective Projection Mathematics

The digital twin renders complex multi-body geometry directly onto an HTML5 Canvas at 60 FPS without external graphical engine bloat. The perspective projection converts 3D world coordinates $[X, Y, Z]^T$ to screen pixels $[X_{\text{screen}}, Y_{\text{screen}}]^T$:

$$\begin{aligned} \mathbf{P}_{\text{cam}} &= \mathbf{R}_x(\text{pitch}) \cdot \mathbf{R}_y(\text{yaw}) \cdot (\mathbf{P}_{\text{world}} + \mathbf{T}_{\text{pan}}) \\ \text{Scale} &= \frac{f_{\text{fov}}}{\max(50, Z_{\text{cam}} + D)} \\ X_{\text{screen}} &= \frac{W}{2} + X_{\text{cam}} \cdot \text{Scale}, \quad Y_{\text{screen}} = \frac{H}{2} - Y_{\text{cam}} \cdot \text{Scale} \end{aligned}$$

The targeting intelligence utilizes a customized single-stage detector optimized for aerial micro-targets (balloons and drones). The neural network combines a cross-stage partial feature extractor with an anchor-free decoupled detection head.



FIGURE 6.0: SINGLE-STAGE ANCHOR-FREE YOLO EDGE DETECTION ARCHITECTURE

Multi-Task Loss Formulation

$$\mathcal{L}_{\text{total}} = \lambda_{\text{box}} \mathcal{L}_{\text{CIoU}} + \lambda_{\text{dfl}} \mathcal{L}_{\text{DFL}} + \lambda_{\text{cls}} \mathcal{L}_{\text{BCE}}$$

Where CIoU optimizes bounding box aspect ratio and centroid Euclidean distance; DFL provides probabilistic boundary regression.

Assembly, Wiring & Safety Protocols

Operating high-energy electro-optical emitters requires rigorous safety fail-safes. The Agesis EYE firmware and ground station implement a five-stage hardware interlock finite state machine.

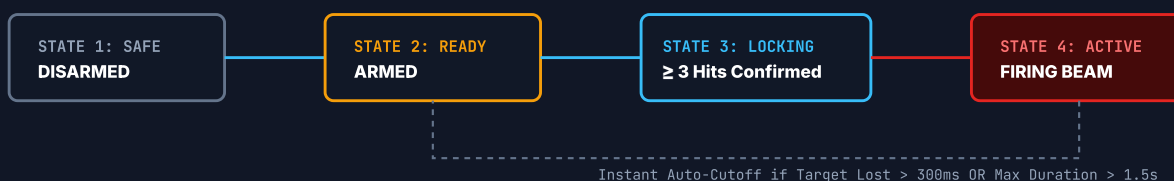


FIGURE 7.0: FIVE-STAGE FAILSAFE INTERLOCK FINITE STATE MACHINE ARCHITECTURE

Safety Operational Checklist

PROTOCOL LEVEL	TRIGGER CONDITION	SYSTEM ACTION
Pre-Arm Interlock	System boot / default state	Emitter hardware gate held LOW (0V). Software locked.
Lock Verification	Detections < 3 consecutive frames	Beam inhibited. Prevents transient glint discharges.
Centering Guard	Boresight error > 16 pixels	Beam inhibited until turret aligns within target cone.
Loss-of-Target Cutoff	Target lost > 300 milliseconds	Instant sub-millisecond hardware disarm.
Thermal Duty Cap	Continuous firing > 1.5 seconds	Mandatory 2.0-second hardware cooldown lockout enforced.